What Offline Fitting Buys Under Benchmark Shift: Predicting LLM Item Outcomes from a Few Labels

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https://github.com/Agrover112/aims-neurips

Abstract

Predicting how a language model answers each item of a new benchmark from a few labelled answers is useful only if what is learned from past benchmarks carries over. We test what carries over with one controlled design: the same models, tasks and label order, changing only where the prior comes from (fixed in advance, tuned on validation, or fitted on training benchmarks), in an online simulation of the competition protocol with three unseen test benchmarks. On benchmarks seen in training, keeping offline estimates of item difficulty gains 0.085 Brier ALC over a Jeffreys Beta-Binomial baseline, nine times the gain of any prior alone. On unseen benchmarks that gain disappears, and only a shrinkage prior transfers: fitted on training it gains 0.014 on test but loses 0.008 on validation. Across the 11 training benchmarks held out one at a time, the gain tracks how far a benchmark's accuracy lies from the prior mean (Spearman 0.93). Gains concentrate at zero to three labels. Item embeddings and length recover no item difficulty on the unseen test benchmarks beyond a random control. A well-fitted Beta-Binomial beats the Rasch models on test, so label sharing across subjects adds nothing to a good prior. We release the simulation, the frozen draws and every saved score.

1 Introduction

Evaluating language models on every item of every benchmark is expensive, so a predictor that infers a model's per-item results from a handful of labelled answers is attractive. The natural recipe is to learn from past benchmarks how items behave and carry that to a new benchmark. Whether it carries is an empirical question, and the answer matters because new benchmarks and new models keep arriving. Learned predictors are known to fail on unfamiliar models while estimators anchored to the observed labels survive [12], and amortized item-response models learn item difficulty from item text [11]. We ask what, among the things one can fit offline, actually transfers to an unseen benchmark.

We use the Predictive AI Evaluation Competition's setting: a *task* is one subject (a model under given evaluation settings) answering one benchmark, labels are bought one at a time, and predictions are scored by Brier ALC at 0, 1, 3, 7, 15 and 31 labels. Our contributions:

- **A controlled comparison.** Beta-Binomial, Rasch and kernel Rasch predictors share tasks, labels and order; only the *prior source* changes (Section 3).
- **A positive control.** Offline fitting must win on benchmarks it has seen; it does, by a wide margin, and the margin vanishes on unseen benchmarks (Section 4).
- **When a transferred prior helps.** A prior fitted on training benchmarks helps on benchmarks whose accuracy is near the prior mean and hurts far from it, shown by leave-one-benchmark-out over the 11 training benchmarks.
- **Open materials.** The frozen draws, saved scores and code to rerun every table and figure.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Track on Evaluations and Datasets.

Related work. Empirical Bayes estimates a prior from related observations; classical shrinkage and small-area estimation show how pooling can stabilize noisy group estimates [1–3]. Prediction-powered inference also combines predictions with observed labels, but targets uncertainty about population quantities rather than sequential item-level prediction [4]. IRT extensions model item–respondent interactions in a latent space or separate language effects from item content [5, 6]. Amortized IRT predicts item parameters from text [11]; our held-out benchmark design asks whether item-level signals and transferred priors remain useful on a new benchmark.

2 Setting and online simulation

We run the online simulation on recorded binary responses from the filtered public data, using disjoint benchmark sets for training, validation, and test (11, 3, and 3 benchmarks). Prior fitting uses training data; validation is used for selecting settings. The competition-shaped draws sample one subject per benchmark in each draw, so a predictor cannot borrow labels from other subjects on that benchmark in the same draw. This is a public-data simulation of the competition protocol.

For each subject–benchmark pair we freeze a split of distinct items into acquisition and evaluation pools, assigning the extra item to acquisition when the count is odd. A frozen random acquisition order supplies nested prefixes of 0, 1, 3, 7, 15, and 31 labels. At each budget the predictor conditions on the allowed prefix and predicts the same evaluation items. Evaluation outcomes are used only to score those predictions. We average Brier scores over query items, then average repeated appearances within each task and give tasks equal weight. Brier ALC combines the six budgets with weights $(0.1, 0.2, 0.2, 0.2, 0.2, 0.1)$. Paired intervals resample whole tasks with their matched predictions. As a separate in-distribution (ID) control, we hold out subjects within training benchmarks and retain questions covered by the fit subjects. The offline comparator can reuse fitted benchmark rates and item difficulties where available; its online update changes ability.

3 Models and prior sources

Beta-Binomial. A task's accuracy π has a Beta prior with mean m and strength s (so $\alpha = ms$, $\beta = (1-m)s$). After n revealed labels with h correct, the posterior is $\mathrm{Beta}(\alpha + h, \beta + n - h)$ and the prediction, the same for every item of the task, is its mean

$$\hat{p} = \frac{ms + h}{s + n}. \tag{1}$$

The Jeffreys prior is $m = 0.5$, $s = 1$ [9]. This is the exact Bayesian sequential update; nothing is learned.

Rasch. The probability that subject s answers item i of benchmark b correctly is

$$\operatorname{logit} p = \mu_b + \theta_s - b_i, \qquad \mu_b \sim \mathcal{N}(m_0, \tau_b^2),\ \theta_s \sim \mathcal{N}(0, \tau_s^2),\ b_i \sim \mathcal{N}(0, \tau_i^2), \tag{2}$$

with benchmark rate μ_b, subject ability θ_s and item difficulty b_i [10]. At every budget we refit the maximum a posteriori (MAP) estimate from scratch on all labels revealed so far in the draw; an unlabelled item gets $b_i = 0$. *Kernel Rasch* replaces the independent item prior by $\mathbf{b} \sim \mathcal{N}(\mathbf{0}, K)$ with $K_{ij} = \tau_i^2[(1-\rho)\delta_{ij} + \rho \exp((\cos(x_i, x_j) - 1)/T)]$ over BGE-M3 item embeddings x_i [8], so similar items share difficulty. With one subject per benchmark, as in the competition-shaped draws, the joint fit equals independent per-task fits (maximum difference 10^{-14} in our check), so Rasch there differs from the Beta-Binomial only in its prior.

Prior sources. Each model is run with three sources of its prior parameters.

- **Fixed in advance:** nothing learned. Beta-Binomial Jeffreys; Rasch $m_0 = 0$ and all widths 1; kernel Rasch additionally $\rho = 0.5$, $T = 0.1$.
- **Tuned on validation:** the best setting on the validation benchmarks from a grid (Beta-Binomial: mean 0.4, strength 1; Rasch and kernel Rasch: the widths and kernel settings selected there).

Table 1: **A prior fitted on training helps on the test benchmarks and hurts on the validation benchmarks; a fixed prior with strong shrinkage does nearly as well on test.** Brier ALC difference to Jeffreys, competition-shaped draws. Test: 3 unseen benchmarks, 45 tasks. Validation: 3 unseen benchmarks, 80 tasks. ID: held-out subjects on the 11 training benchmarks, 114 tasks, where the training-fitted Beta-Binomial prior is refit without those subjects.

Prior source	Model	Test (OOD)	Validation (OOD)	ID
Fixed in advance	Beta-Binomial (Jeffreys)	reference	reference	reference
	Rasch	-0.0102 $[-0.0118, -0.0085]$	$+0.0031$ $[-0.0005, +0.0066]$	-0.0073 $[-0.0091, -0.0055]$
	Kernel Rasch	-0.0105 $[-0.0121, -0.0088]$	$+0.0029$ $[-0.0008, +0.0065]$	-0.0072 $[-0.0087, -0.0058]$
Tuned on validation	Beta-Binomial	$+0.0055$ $[+0.0038, +0.0072]$	-0.0015 $[-0.0040, +0.0010]$	$+0.0041$ $[+0.0027, +0.0055]$
	Rasch	-0.0032 $[-0.0035, -0.0029]$	$+0.0004$ $[-0.0006, +0.0014]$	-0.0024 $[-0.0028, -0.0019]$
	Kernel Rasch	-0.0024 $[-0.0027, -0.0021]$	-0.0003 $[-0.0010, +0.0004]$	$+0.0001$ $[-0.0004, +0.0005]$
Fitted on training	Beta-Binomial	-0.0138 $[-0.0158, -0.0116]$	$+0.0079$ $[+0.0024, +0.0136]$	-0.0090 $[-0.0115, -0.0063]$
	Rasch	-0.0119 $[-0.0133, -0.0106]$	$+0.0052$ $[+0.0008, +0.0097]$	-0.0081 $[-0.0101, -0.0060]$
	Kernel Rasch	-0.0118 $[-0.0131, -0.0105]$	$+0.0036$ $[-0.0007, +0.0076]$	-0.0063 $[-0.0076, -0.0049]$

- **Fitted on training:** the 11 training benchmarks only. The Beta-Binomial prior maximises the Beta-Binomial marginal likelihood of the training tasks, weighting benchmarks equally (mean 0.522, strength 3.68). The Rasch priors come from a joint MAP fit to 4.34 million training responses (4,809 subjects, 109,987 items) with empirical-Bayes hyperparameters averaged per benchmark so that the benchmark with most subjects does not dominate: $m_0 = 0.175$, $\tau_b = 0.75$, $\tau_s = 1.38$, $\tau_i = 1.49$. Only these numbers transfer; the fitted μ_b, θ_s and b_i belong to training benchmarks, subjects and items.

Offline model that keeps its fitted values. For the ID control, the offline Rasch keeps its fitted μ_b and b_i for a benchmark and items it has seen and updates only the subject ability θ_s online (a one-dimensional MAP on the task's own labels). For an unseen benchmark or item it falls back to m_0 and 0, i.e. to the fitted prior alone.

4 Results

All differences are Brier ALC minus the Jeffreys baseline (negative is better), with paired 95% bootstrap intervals over tasks.

4.1 The source of the prior

Table 1 compares the nine combinations. On test, a Beta-Binomial with a prior fitted on training is the best predictor (-0.0138 $[-0.0158, -0.0116]$), four times the gain of the validation-tuned Rasch, and better than any Rasch model (fitted Rasch minus fitted Beta-Binomial $+0.0019$ $[+0.0007, +0.0031]$). The validation-tuned Beta-Binomial prior is *worse* than Jeffreys on test ($+0.0055$) though it is the best Beta-Binomial on validation. Every comparison between prior sources changes sign between test and validation (Appendix B); fresh draws change no result by more than 0.0007 (Appendix C).

4.2 Offline fitting pays off where its fitted values apply

On the ID panel the offline model that keeps its fitted values scores -0.0852 $[-0.0955, -0.0747]$ against -0.0090 for the fitted Beta-Binomial prior and -0.0081 for the fitted Rasch prior (Figure 2); its item difficulties alone, at budget zero, reach a within-task AUC of 0.836. On the unseen test benchmarks it equals the fitted prior (-0.0115 against -0.0119): nothing it knows about particular items or benchmark rates is available. This is the positive control the design requires, and it separates

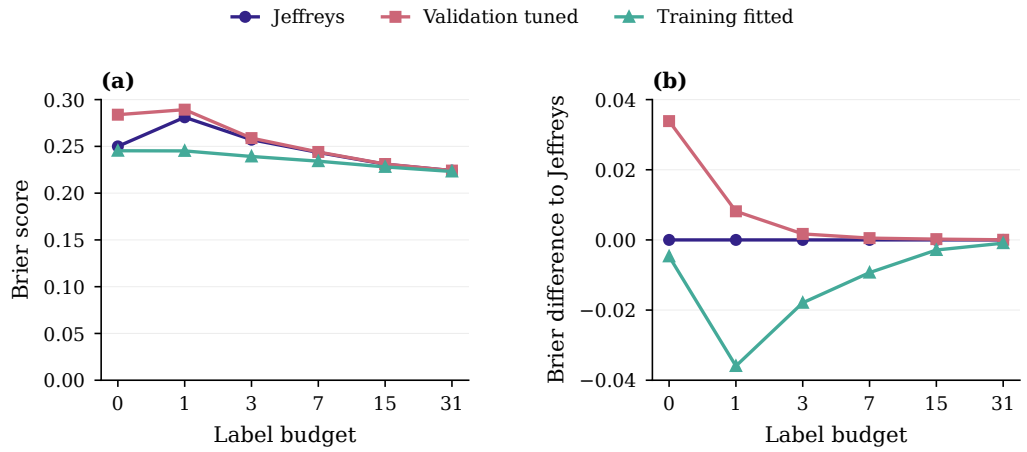


Figure 1: **The prior's source matters at zero to three labels and has faded by 31.** Brier score (a) and its difference to Jeffreys (b) per label budget on the three unseen test benchmarks, competition-shaped draws, for the Beta-Binomial with a fixed (Jeffreys, solid), validation-tuned (dashed) and training-fitted (dotted) prior. Jeffreys jumps to 0.75 or 0.25 after one answer, so the gap is largest at one label; at 31 labels the three differ by less than 0.001.

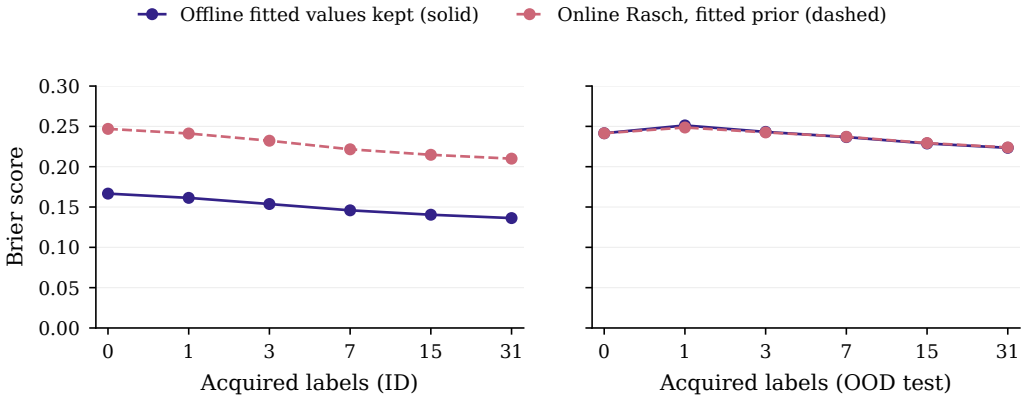


Figure 2: **Offline estimates win on familiar benchmarks and have no edge on unfamiliar ones.** Brier score per label budget for the offline Rasch keeping its fitted benchmark rates and item difficulties and updating only ability (solid), against the online Rasch with the same fitted prior (dashed). Left: ID, held-out subjects on the 11 training benchmarks. Right: OOD test benchmarks, where the offline model has no fitted values and falls back to the prior.

two things that are easy to conflate: item-level estimates transfer only to the items they were fitted on, while the prior's spreads carry over to new benchmarks.

4.3 When does a fitted prior help?

Figure 3 shows the test and validation benchmarks separately: all three test benchmarks agree (-0.013 to -0.015), and on validation `ai2d_test` agrees while `bird_sql` goes the other way ($+0.025$). Holding out each training benchmark in turn (Figure 4, Appendix D) the fitted prior significantly helps on 5 of 11, significantly hurts on none, and the macro mean gain is -0.0056. What separates the benchmarks is not whether they lie inside the range of training benchmark rates (`alignment_faking` and `harmbench` lose while inside it) but how far their task accuracies sit from

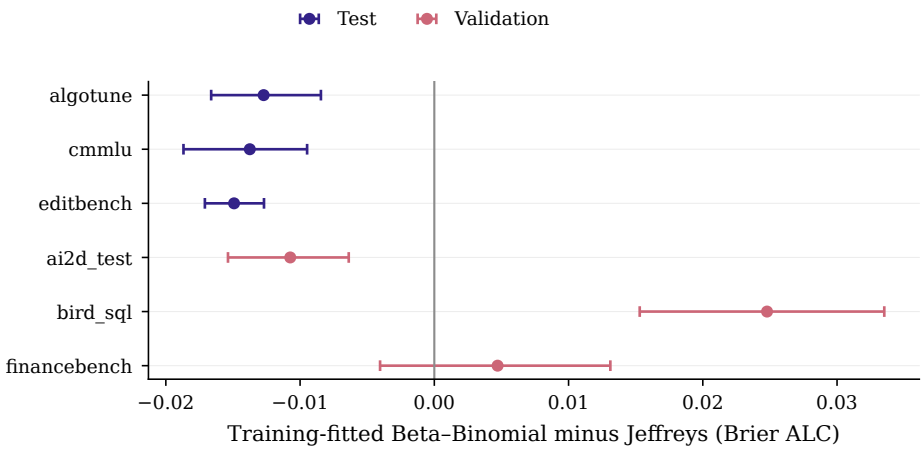


Figure 3: **The training-fitted prior helps on all three test benchmarks and on one validation benchmark, and hurts on the hardest.** Brier ALC difference between the training-fitted Beta-Binomial and Jeffreys for each unseen benchmark, with 95% intervals over tasks. The validation benchmark `bird_sql` has task accuracy 0.12, far below every training benchmark.

Table 2: Item-difficulty transfer (WP7). Cells report within-benchmark Spearman correlation and pairwise accuracy, respectively.

Feature	ID (11)	Validation (3)	Test (3)
Embedding	0.400 / 0.649	0.196 / 0.570	−0.020 / 0.494
Log character count	0.107 / 0.539	0.058 / 0.525	0.061 / 0.521
Random score	−0.003 / 0.499	−0.024 / 0.490	0.076 / 0.529

the prior mean: the Spearman correlation between the gain and that distance is 0.93 over the 11 benchmarks, against 0.01 for the benchmark's mean accuracy. Shrinking toward 0.52 helps tasks near 0.52 and hurts tasks far from it. The three test benchmarks (mean accuracy 0.54–0.68) sit near the prior mean, which is why test looks good. We found this pattern after seeing the results, so it is an explanation and not a predictor.

Under the fitted priors, the simplest Rasch model is also the best on test and the worst on validation: with benchmark rate only the gain is −0.0172 on test and +0.0211 on validation (Appendix A). The subject-ability term is what lets a benchmark far from the prior move away from it, at a cost in variance when the benchmark is near it. The kernel adds nothing on test.

5 Item-difficulty transfer

We ask whether item features recover the ordering of observed item difficulty. The target is the mean binary correctness per item, centered within benchmark for fitting. We compare a ridge probe of item embeddings, a probe using log character count, and a sampled random score. The ID diagnostic holds out items within the 11 training benchmarks. The other two columns evaluate entire unseen benchmarks. All methods use matched items and sampled pairs. Pairwise accuracy excludes observed ties and gives half credit for predicted ties. Results are averaged across benchmarks.

Embeddings show a clear signal on known benchmarks. The historical figure of about 0.09 Spearman (0.53 pairwise) pools validation and test across all six unseen benchmarks. On the three test benchmarks alone, neither feature beats the matched sampled random control on either metric. This is a descriptive result: the sampled control can fluctuate around chance, and three benchmarks cannot establish that the features have no transfer value.

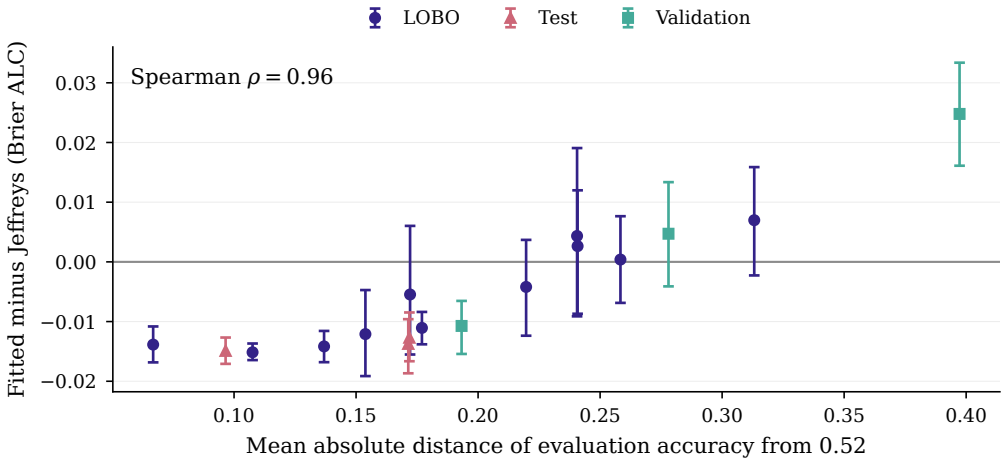


Figure 4: **A fitted prior helps when a benchmark's accuracy is near the prior mean.** Each point is one of 17 benchmarks. For each of the 11 training benchmarks, the prior was refit without any of that benchmark's rows and scored on held-out subjects. The six validation and test benchmarks use the original training-only prior. The horizontal axis uses task accuracy on the evaluation items; the vertical axis shows fitted minus Jeffreys Brier ALC, with 95% task-bootstrap intervals. This is an explanatory analysis using observed outcomes.

6 Limitations

The filtered public data covers only three test benchmarks in the online simulation. Multiple draws and acquisition orders measure variation within this population; they do not supply independent tests of transfer to new benchmarks. Task bootstrap intervals likewise do not capture uncertainty from sampling new benchmarks. Results apply to the eligible response panels and random label acquisition, and need not carry over to other benchmark mixes or selection policies. The 17-benchmark sensitivity analysis reuses held-out-subject tasks for the 11 original training benchmarks. The test triples overlap, so their distribution describes sensitivity within this dataset rather than independent estimates of transfer. The ID comparison has item response history that is unavailable under benchmark cold start. Observed item difficulty also depends on which subjects answered the item. The leave-one-benchmark-out association uses realized outcomes to explain when the fitted prior helps; it is not a way to select a prior before observing a new benchmark.

Further limits of our own. The test benchmarks are close to the prior mean, so the size of the test gain depends on this split; with a different draw of three benchmarks the sign could differ, as it does on validation. Test scores were inspected repeatedly while building the models, so the test analyses are exploratory: all selection used validation, but test was not read once. The kernel settings were chosen on validation and not on the ID panel. We did not rerun the subject-metadata prior in this design, and we dropped reasoning-trace features, which are not available at test time.

7 Conclusion

What transfers from offline fitting to an unseen benchmark is a shrinkage prior, not item-level estimates. Offline fitting is worth a very large gain where its fitted values apply and none where they do not. The prior helps when the new benchmark's accuracy is near the prior mean and hurts when far from it, so the same fitted prior looked good on test and bad on validation. A well-fitted Beta-Binomial is the strongest estimator in our test; label sharing across subjects added nothing to it.

Disclosure and rerun instructions

Training data. All fitted quantities (the offline Rasch fit, the Beta-Binomial empirical-Bayes prior, the item probes) use only the responses of the 11 training benchmarks in the competition's measurement database after the filtering described in Section 2; no other response data is used. **Pretrained models.** Item text is embedded with the frozen BGE-M3 model; nothing else is pretrained or fine-tuned. **LLM coding assistants.** Claude Code (Anthropic) assisted with writing code, analysis scripts and drafting text; every reported number was produced by running the code in the repository, and the authors checked the numbers. OpenAI Codex assisted with the reviewer analyses, figures, and text revisions.

The shared handoff contains frozen draws, scores, and competition-shaped test predictions. The GitHub analysis branch contains online-simulation and analysis code, aggregate results, and figure PDFs. Keep private response and prediction files in the authorized shared storage. The command `python organized/testing/eandd/wp9.py` with `-handoff` pointing to the local handoff and `-output` pointing to a fresh directory rebuilds the figures and checks the saved Brier scores against the saved predictions. The analysis README identifies the handoff location and documents how to render figures from aggregate CSVs alone. The WP7 item results are in `paper/eandd/data/wp7-summary.csv`. Record the analysis commit and retain the frozen draws for any rerun.

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A Rasch terms under the fitted priors

Table 3: Brier ALC difference to Jeffreys for Rasch with terms switched off (prior width set near zero), under the training-fitted priors. Competition-shaped draws on test and validation, and multi-pair draws on test.

Model	Test	Validation	Multi-pair test
Benchmark rate only	-0.0172 [-0.0202, -0.0142]	$+0.0211$ [$+0.0114$, $+0.0311$]	-0.0153 [-0.0186, -0.0118]
+ subject ability	-0.0089 [-0.0100, -0.0080]	$+0.0029$ [-0.0003, $+0.0060$]	-0.0092 [-0.0103, -0.0081]
+ item difficulty (Rasch)	-0.0119 [-0.0133, -0.0106]	$+0.0052$ [$+0.0007$, $+0.0098$]	-0.0124 [-0.0139, -0.0110]
+ kernel	-0.0118 [-0.0131, -0.0105]	$+0.0036$ [-0.0006, $+0.0078$]	-0.0125 [-0.0140, -0.0111]
Rate and item, no ability	-0.0174 [-0.0208, -0.0139]	$+0.0282$ [$+0.0164$, $+0.0401$]	-0.0158 [-0.0196, -0.0119]
Ability and item, no rate	-0.0140 [-0.0157, -0.0122]	$+0.0078$ [$+0.0022$, $+0.0135$]	-0.0144 [-0.0162, -0.0126]

B Paired comparisons between prior sources

Table 4: Paired Brier ALC differences between prior sources, with 95% intervals over tasks. Each comparison changes sign between test and validation.

Comparison (a − b)	Test	Validation	ID
Rasch fitted − Rasch fixed	-0.0017 [-0.0028, -0.0005]	$+0.0021$ [$+0.0006$, $+0.0036$]	–
Kernel fitted − Rasch fitted	$+0.0001$ [-0.0002, $+0.0004$]	-0.0016 [-0.0021, -0.0012]	–
Rasch fitted − Beta-Binomial fitted	$+0.0019$ [$+0.0007$, $+0.0031$]	-0.0027 [-0.0041, -0.0012]	$+0.0009$ [-0.0001, $+0.0018$]

C Draw-seed robustness

Table 5: Brier ALC difference to Jeffreys on three fresh sets of 150 competition-shaped draws per role (seeds 20261201–03); results on the frozen draws are in Table 1.

Model	Test	Validation
Beta-Binomial fitted	-0.0137 / -0.0144 / -0.0139	$+0.0078$ / $+0.0083$ / $+0.0089$
Rasch fitted	-0.0120 / -0.0124 / -0.0123	$+0.0051$ / $+0.0054$ / $+0.0060$
Beta-Binomial tuned	$+0.0060$ / $+0.0059$ / $+0.0063$	-0.0015 / -0.0016 / -0.0018

D Leave-one-benchmark-out for the fitted prior

Table 6: Fitted Beta-Binomial prior (refit without the held-out benchmark) minus Jeffreys, on each training benchmark's held-out-subject tasks, with the mean distance of its task accuracies from the prior mean 0.52.

Held-out benchmark	Tasks	Gain [95% CI]	Distance from 0.52
`hallusionbench`	20	-0.0151 [-0.0165, -0.0136]	0.11
`bridging_gap`	20	-0.0142 [-0.0167, -0.0114]	0.14
`helm_afr`	3	-0.0139 [-0.0168, -0.0108]	0.07
`afrimedqa`	6	-0.0121 [-0.0189, -0.0046]	0.15
`afrieval`	20	-0.0111 [-0.0138, -0.0085]	0.18
`cruxeval`	5	-0.0055 [-0.0155, $+0.0056$]	0.17
`brace`	6	-0.0042 [-0.0121, $+0.0037$]	0.22
`agi_elo`	4	$+0.0004$ [-0.0069, $+0.0076$]	0.26
`appworld`	4	$+0.0026$ [-0.0087, $+0.0120$]	0.24
`alignment_faking`	20	$+0.0043$ [-0.0092, $+0.0188$]	0.24
`harmbench`	6	$+0.0070$ [-0.0023, $+0.0161$]	0.31

E Sensitivity to the benchmark split

We held out each of the 680 triples, refitted the benchmark-weighted empirical Beta prior on the other 14 benchmarks, and rescored the frozen trajectories. The mean macro difference was -0.0037, with 5th–95th percentiles [-0.0095, $+0.0099$]. The fitted prior hurt on 17.5% of splits. The original test benchmark names gave -0.0087 (percentile 13) under refitting. This refit includes the former validation benchmarks among the 14 eligible fit benchmarks for the original test triple, so its result differs from the original 11-benchmark fit. The 11 original training benchmarks are scored on held-out-subject tasks from the existing leave-one-out panel. These overlapping triples describe split sensitivity within this dataset.

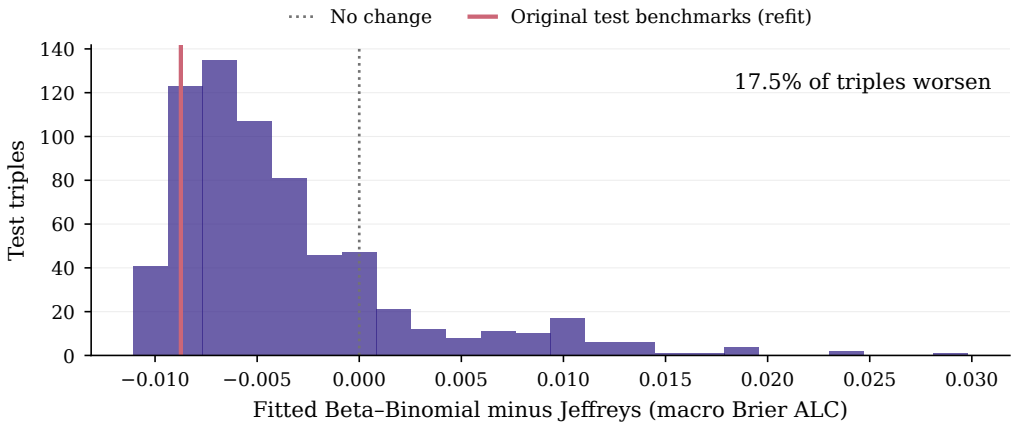


Figure 5: Fitted Beta-Binomial minus Jeffreys over all 680 triples. Each empirical prior is fitted on the 14 benchmarks outside its triple. The vertical line marks the original test benchmark names under this refitting protocol.

F Alternative evaluation metrics

Table 7: Competition-shaped replay. Brier, log loss, and ten-bin task ECE are averaged using the official budget weights; lower is better. AUC uses within-task item rankings at 31 labels, where higher is better. Dashes indicate no informative ranking; a constant prediction ties all items. Per-budget AUC and alternative weights are reported in the accompanying CSVs.

Split	Model	Brier	Log loss	ECE	AUC at 31
Test	Jeffreys	0.2499	0.7045	0.1426	–
Test	Fitted Beta-Binomial	0.2362	0.6656	0.1113	–
Test	Fitted Rasch	0.2380	0.6712	0.1164	–
Test	Fitted kernel Rasch	0.2382	0.6717	0.1230	0.571
Validation	Jeffreys	0.1738	0.5159	0.1405	–
Validation	Fitted Beta-Binomial	0.1817	0.5404	0.1677	–
Validation	Fitted Rasch	0.1790	0.5323	0.1580	–
Validation	Fitted kernel Rasch	0.1774	0.5279	0.1603	0.618
ID	Jeffreys	0.2357	0.6681	0.1407	–
ID	Fitted Beta-Binomial	0.2268	0.6437	0.1235	–
ID	Fitted Rasch	0.2276	0.6460	0.1245	–
ID	Fitted kernel Rasch	0.2295	0.6506	0.1336	0.581
ID	Offline Rasch	0.1506	0.4634	0.1161	0.836